

Politecnico di Torino  
Bachelor Degree in Mechanical/Automotive Engineering

Academic Year 2025-2026

01NLMJM, 01NLMLI Science and technology of materials/Technology of metallic materials

## Science and technology of materials

**Prof. Francesco Baino**

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# PROGRAM OF THIS COURSE:

- The aim of this course is to provide Mechanical and Automotive Engineering students basic knowledge on different classes of materials (**metals, polymers, ceramics, glasses and composites**) **and their properties** (in particular thermal and mechanical).
- Background: Mathematics, Chemistry, Physics

# PROGRAM OF THIS COURSE:

- Science and technology of materials (50 h)  
(Francesco Baino, [francesco.baino@polito.it](mailto:francesco.baino@polito.it))
- Technology of metallic materials (50 h)  
(Giovanni Maizza, [giovanni.maizza@polito.it](mailto:giovanni.maizza@polito.it))

# Science and technology of materials

- LECTURES: about 35 HOURS

Prof. Francesco Baino

- EXERCISES: about 15 HOURS

Dr. Stefano De la Pierre, [stefano.delapierre@polito.it](mailto:stefano.delapierre@polito.it)

Dr. Sara Ferraris, [sara.ferraris@polito.it](mailto:sara.ferraris@polito.it)

Dr. Mariacristina Martorino, [mariacristina.martorino@polito.it](mailto:mariacristina.martorino@polito.it)

# Timetable

| Monday | Tuesday                              | Wednesday                              | Thursday                             | Friday                                                                                                                                                 | Saturday | Sunday |
|--------|--------------------------------------|----------------------------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| -----  | -----                                | -----                                  | -----                                | -----                                                                                                                                                  | -----    | -----  |
|        |                                      |                                        | 1,5 h, ExC<br>08:30-10:00<br>Room 4D |                                                                                                                                                        |          |        |
|        |                                      | 1,5 h, Lect.<br>10:00-11:30<br>Room 1B | 3 h, Lect.<br>10:00-13:00<br>Room 8C | <div> <p>There might be <b>some variations:</b><br/>please look carefully at the section «<b>Advices</b>» on the <b>STM course webpage!</b></p> </div> |          |        |
|        | 3 h, Lect.<br>13:00-16:00<br>Room 14 |                                        | 1,5 h, ExD<br>13:00-14:30<br>Room 4C |                                                                                                                                                        |          |        |
|        | 1,5 h, ExA<br>16:00-17:30<br>Room 9S |                                        |                                      |                                                                                                                                                        |          |        |
|        | 1,5 h, ExB<br>17:30-19:00<br>Room 9S |                                        |                                      |                                                                                                                                                        |          |        |



Details will be written on the course webpage every week: don't worry!

# Text Book and other material

- **Materials Science and Engineering**  
**W.D. Callister (EdiSES)**

- Your notes.
- Other books on Materials Science
- Slides of the course, which are uploaded weekly on the course webpage

# Information about the STM exam

(see also the course webpage for more details)

- The exam is a written one (duration: around 45 min -1 hour) for STM, based on what done during lectures and exercise classes
- The exam will comprise open questions needing a concise but exhaustive response, multiple choice questions, and exercises. The total score will be 32 (30 cum laude)
- The final grade for this integrated STM-TMM course is ONE (average of the two STM and TMM grades)
- STM grade is valid indefinitely until you pass TMM.
- **Warning #1:** once you have passed the STM exam (score  $\geq 18$ ), it will be considered automatically accepted unless you reject it by e-mail by 48 h from publication.
- **Warning #2:** you **must always book for the STM exam** even if you already passed STM and have only to give TMM!

# Assistance

- write to me or to Stefano De La Pierre, Sara Ferraris, Mariacristina Martorino (about exercises)
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